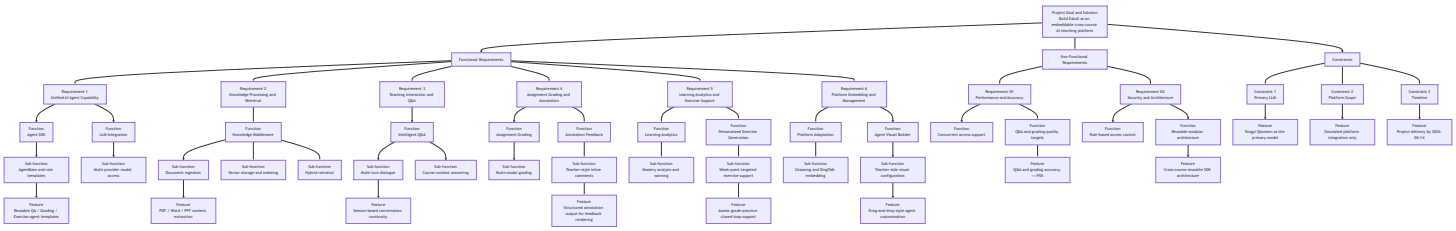
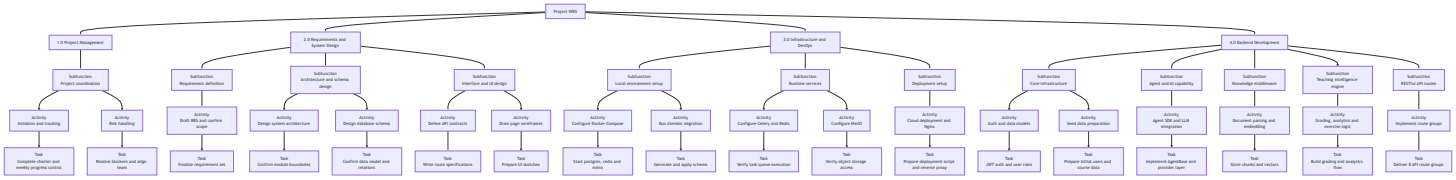
EduAI RBS / WBS / Schedule Diagrams

本文件用于重绘项目的 `RBS` 、 `WBS` 和 `Schedule` 图示，面向课程作业与阶段汇报使用。

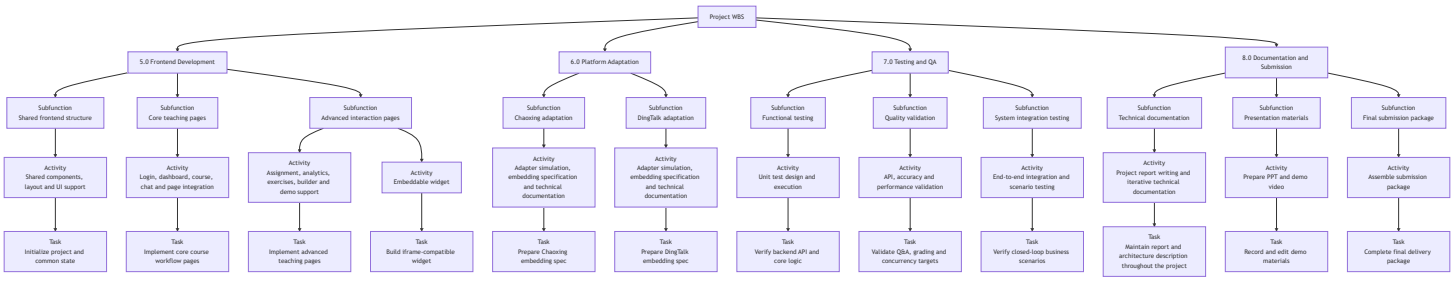
1. Requirement Breakdown Structure (RBS)



2. Work Breakdown Structure (WBS) - Part 1



3. Work Breakdown Structure (WBS) - Part 2



4. Estimated Working Hours

4.1 Module-Level Summary

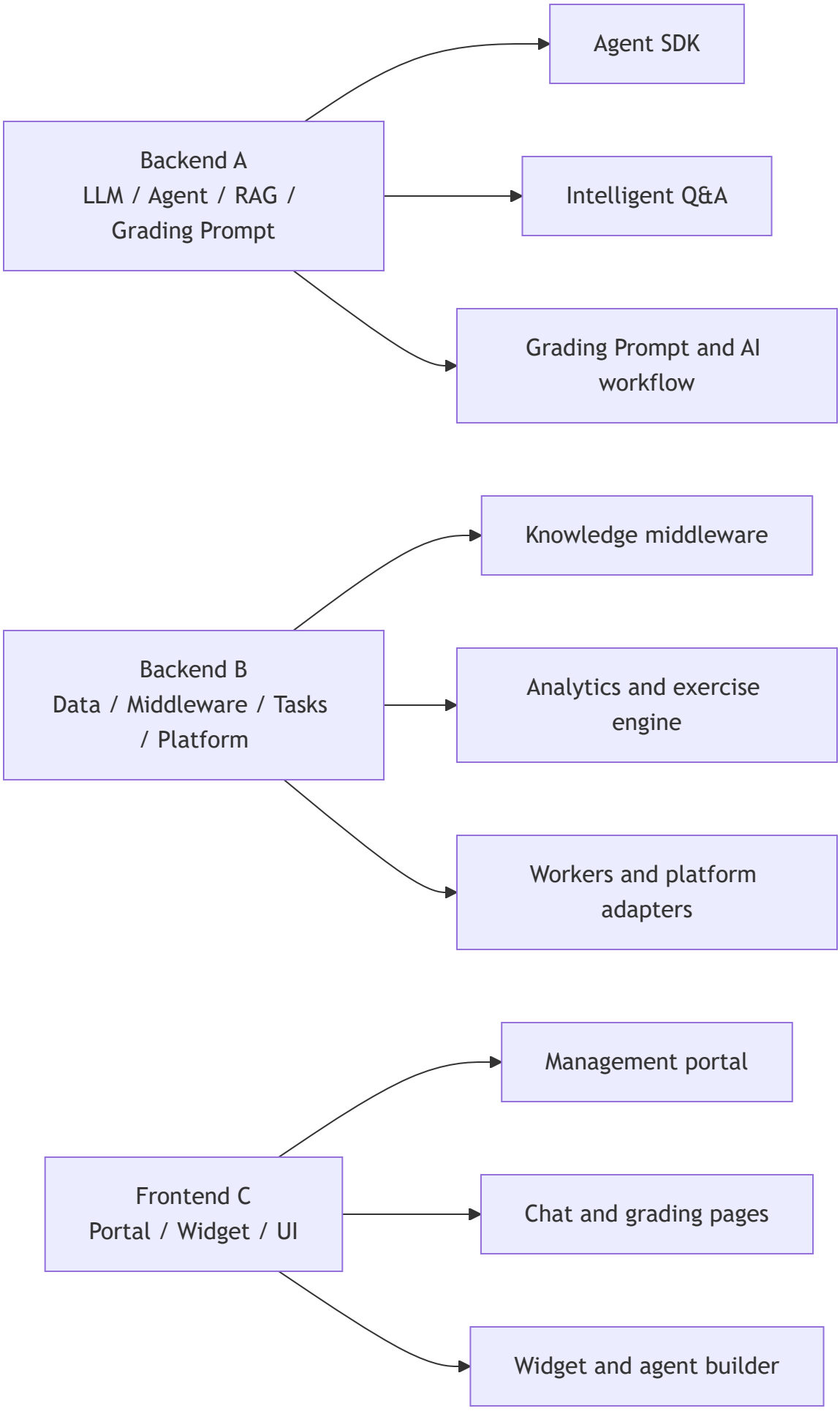
WBS	Module	Estimated Hours	Primary Responsible
1.0	Project Management	24	All members
2.0	Requirements and System Design	40	All members
3.0	Infrastructure and DevOps	32	Backend A / Backend B
4.0	Backend Development	120	Backend A / Backend B
5.0	Frontend Development	96	Frontend C / Integrated D
6.0	Platform Adaptation	16	Integrated D / Backend B
7.0	Testing and QA	40	Integrated D (lead) / All members
8.0	Documentation and Submission	32	Integrated D / All members
	Total Working Hours	**400**	

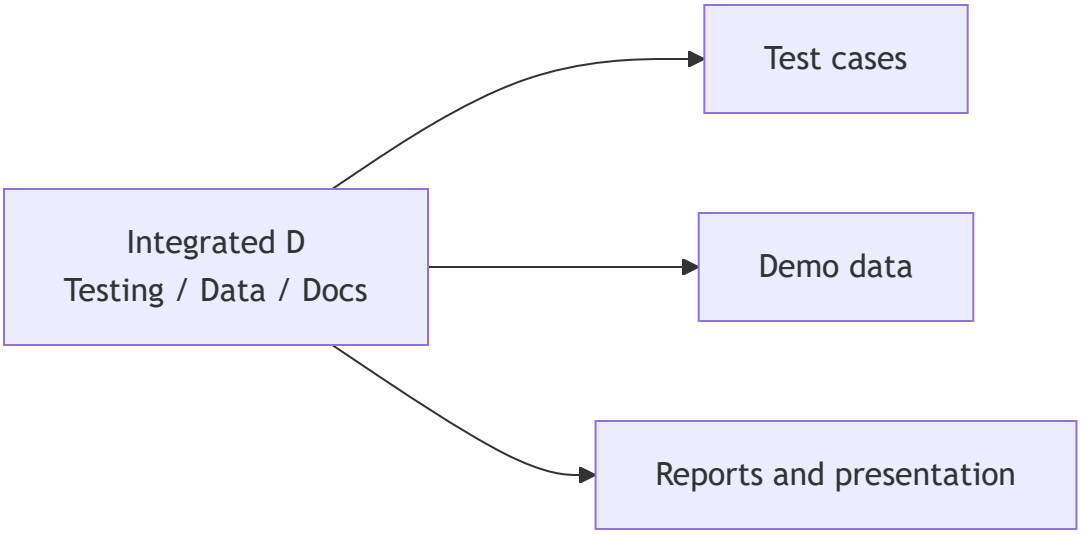
4.2 Task-Level Estimation for Scheduling

WBS	Task / Work Package	Estimated Hours	Suggested Owner
1.1	Complete charter and weekly progress control	16	All members
1.2	Resolve blockers and align team	8	All members
2.1	Finalize requirement set	8	All members
2.2	Confirm module boundaries	12	All members
2.3	Confirm data model and relations	8	Backend A / Backend B

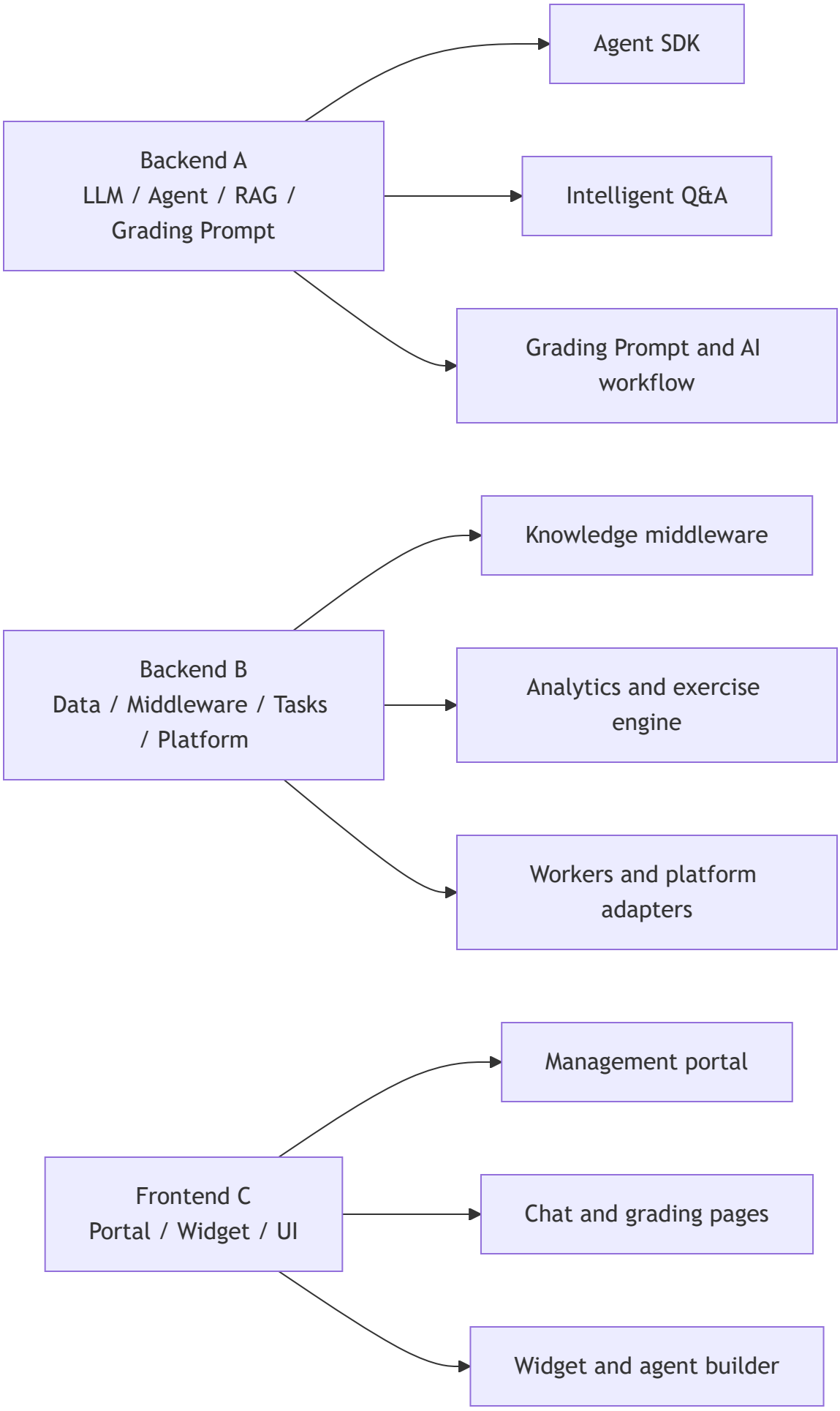
WBS	Task / Work Package	Estimated Hours	Suggested Owner
2.4	Write route specifications	8	Backend A / Backend B
2.5	Prepare UI sketches	4	Frontend C
3.1	Start postgres, redis and minio	8	Backend A
3.2	Generate and apply schema	4	Backend A
3.3	Verify task queue execution	8	Backend B
3.4	Verify object storage access	4	Backend B
3.5	Prepare deployment script and reverse proxy	8	Backend A / Backend B
4.1	JWT auth and user roles	10	Backend A
4.2	Prepare initial users and course data	10	Backend B
4.3	Implement AgentBase and provider layer	16	Backend A
4.4	Implement QA / grading / exercise agent capability	16	Backend A
4.5	Store chunks and vectors	12	Backend B
4.6	Build grading and analytics flow	16	Backend A / Backend B
4.7	Deliver Auth / Courses APIs	6	Backend B
4.8	Deliver Agents / Assignments APIs	6	Backend A / Backend B
4.9	Deliver Chat / Analytics APIs	6	Backend A / Backend B
4.10	Deliver Exercises / Platform APIs	6	Backend B
4.11	Resource upload and async processing loop	8	Backend B
4.12	Q&A closed-loop integration	8	Backend A

WBS	Task / Work Package	Estimated Hours	Suggested Owner
5.1	Initialize project and common state	8	Frontend C / Integrated D
5.2	Implement login, dashboard, course and chat pages	32	Frontend C / Integrated D
5.3	Implement assignment, analytics, exercises and builder pages	40	Frontend C
5.4	Build iframe-compatible widget	16	Frontend C / Integrated D
6.1	Prepare Chaoxing embedding spec	8	Integrated D
6.2	Prepare DingTalk embedding spec	8	Integrated D
7.1	Unit test design and execution for backend/API logic	14	Integrated D
7.2	API, Q&A, grading and concurrency validation	14	Integrated D
7.3	End-to-end integration and closed-loop scenario verification	12	All members
8.1	Project report writing and iterative update	12	All members
8.2	Record and edit demo materials	8	Integrated D
8.3	Complete final delivery package	12	Integrated D


Backend A
LLM / Agent / RAG / Grading Prompt
Agent SDK
Intelligent Q&A
Grading Prompt and AI workflow
Backend B
Data / Middleware / Tasks / Platform
Knowledge middleware
Analytics and exercise engine
Workers and platform adapters
Frontend C
Portal / Widget / UI
Management portal
Chat and grading pages
Widget and agent builder



5. Responsibility Mapping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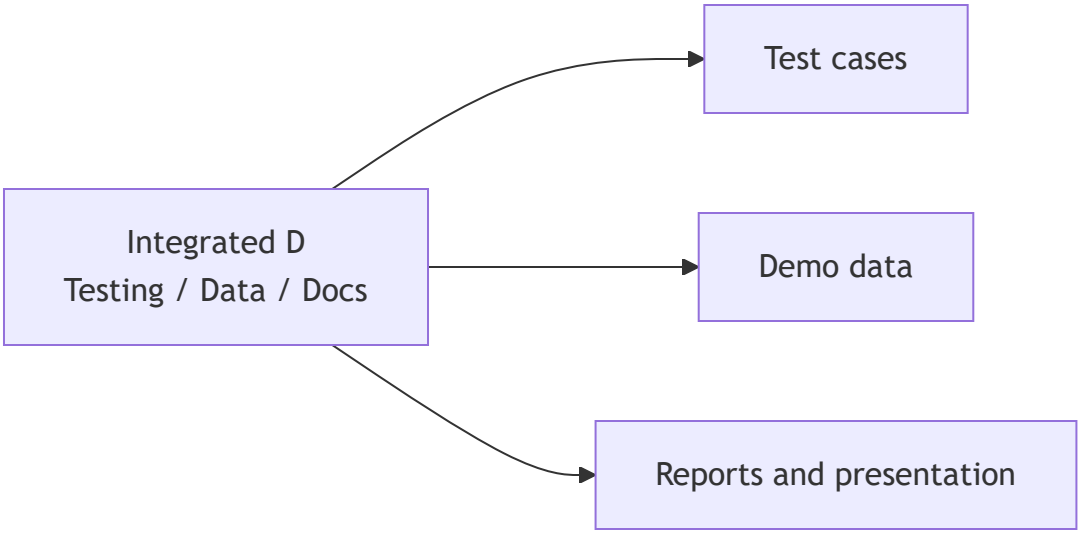
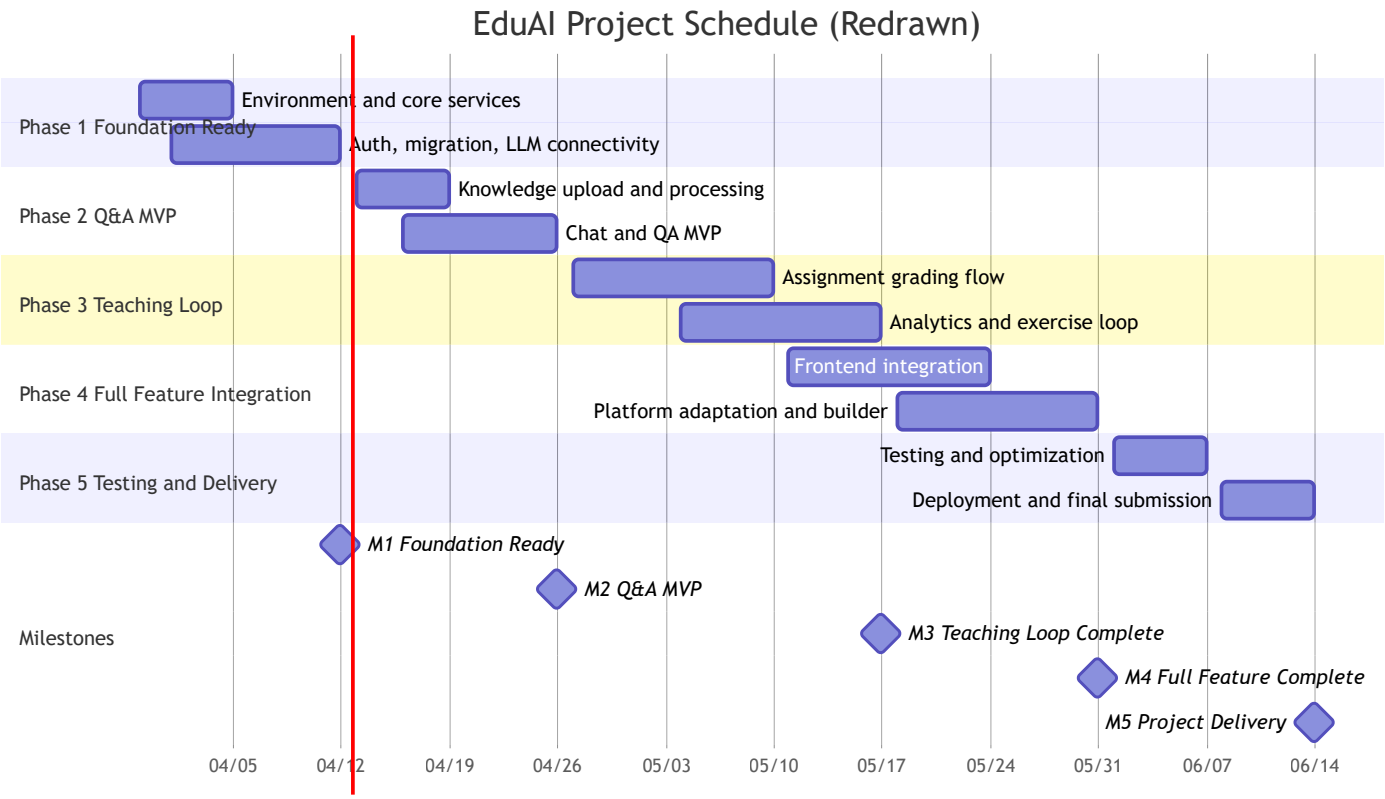
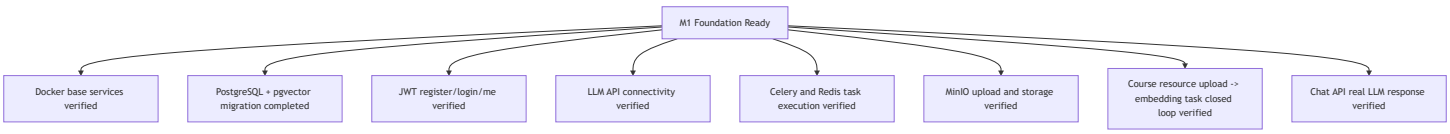


6. Project Schedule and Milestones



7. Current Progress Alignment